

```
!pip install transformers datasets evaluate torch matplotlib
```

Defaulting to user installation because normal site-packages is not writeable

Requirement already satisfied: transformers in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (4.49.0)

Requirement already satisfied: datasets in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (3.3.2)

Requirement already satisfied: evaluate in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (0.4.3)

Requirement already satisfied: torch in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (2.6.0)

Requirement already satisfied: matplotlib in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (3.10.1)

Requirement already satisfied: filelock in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from transformers) (3.17.0)

Requirement already satisfied: huggingface-hub<1.0,>=0.26.0 in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from transformers) (0.29.2)

Requirement already satisfied: numpy>=1.17 in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from transformers) (2.2.3)

Requirement already satisfied: packaging>=20.0 in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from transformers) (24.2)

Requirement already satisfied: pyyaml>=5.1 in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from transformers) (6.0.2)

Requirement already satisfied: regex!=2019.12.17 in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from transformers) (2024.11.6)

Requirement already satisfied: requests in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from transformers) (2.32.3)

Requirement already satisfied: tokenizers<0.22,>=0.21 in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from transformers) (0.21.0)

Requirement already satisfied: safetensors>=0.4.1 in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from transformers) (0.5.3)

Requirement already satisfied: tqdm>=4.27 in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from transformers) (4.67.1)

Requirement already satisfied: pyarrow>=15.0.0 in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from datasets) (19.0.1)

Requirement already satisfied: dill<0.3.9,>=0.3.0 in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from

datasets) (0.3.8)
Requirement already satisfied: pandas in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from datasets) (2.2.3)
Requirement already satisfied: xxhash in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from datasets) (3.5.0)
Requirement already satisfied: multiprocess<0.70.17 in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from datasets) (0.70.16)
Requirement already satisfied: fsspec<=2024.12.0,>=2023.1.0 in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from fsspec[http]<=2024.12.0,>=2023.1.0->datasets) (2024.12.0)
Requirement already satisfied: aiohttp in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from datasets) (3.11.13)
Requirement already satisfied: typing-extensions>=4.10.0 in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from torch) (4.12.2)
Requirement already satisfied: networkx in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from torch) (3.4.2)
Requirement already satisfied: jinja2 in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from torch) (3.1.6)
Requirement already satisfied: setuptools in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from torch) (75.8.2)
Requirement already satisfied: sympy==1.13.1 in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from torch) (1.13.1)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from sympy==1.13.1->torch) (1.3.0)
Requirement already satisfied: contourpy>=1.0.1 in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from matplotlib) (1.3.1)
Requirement already satisfied: cycler>=0.10 in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from matplotlib) (4.56.0)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from matplotlib) (1.4.8)
Requirement already satisfied: pillow>=8 in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from matplotlib) (11.1.0)
Requirement already satisfied: pyparsing>=2.3.1 in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from matplotlib) (3.2.1)
Requirement already satisfied: python-dateutil>=2.7 in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from

matplotlib) (2.9.0.post0)
Requirement already satisfied: aiohappyeyeballs>=2.3.0 in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from aiohttp->datasets) (2.5.0)
Requirement already satisfied: aiosignal>=1.1.2 in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from aiohttp->datasets) (1.3.2)
Requirement already satisfied: attrs>=17.3.0 in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from aiohttp->datasets) (25.1.0)
Requirement already satisfied: frozenlist>=1.1.1 in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from aiohttp->datasets) (1.5.0)
Requirement already satisfied: multidict<7.0,>=4.5 in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from aiohttp->datasets) (6.1.0)
Requirement already satisfied: propcache>=0.2.0 in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from aiohttp->datasets) (0.3.0)
Requirement already satisfied: yarl<2.0,>=1.17.0 in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from aiohttp->datasets) (1.18.3)
Requirement already satisfied: six>=1.5 in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from python-dateutil>=2.7->matplotlib) (1.17.0)
Requirement already satisfied: charset-normalizer<4,>=2 in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from requests->transformers) (3.4.1)
Requirement already satisfied: idna<4,>=2.5 in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from requests->transformers) (3.10)
Requirement already satisfied: urllib3<3,>=1.21.1 in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from requests->transformers) (2.3.0)
Requirement already satisfied: certifi>=2017.4.17 in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from requests->transformers) (2025.1.31)
Requirement already satisfied: colorama in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from tqdm>=4.27->transformers) (0.4.6)
Requirement already satisfied: MarkupSafe>=2.0 in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from jinja2->torch) (3.0.2)
Requirement already satisfied: pytz>=2020.1 in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from pandas->datasets) (2025.1)
Requirement already satisfied: tzdata>=2022.7 in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from pandas->datasets) (2025.1)

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[notice] A new release of pip is available: 24.3.1 -> 25.0.1
[notice] To update, run: C:\Program Files\Python312\python.exe -m pip
install --upgrade pip
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```
pip install pandas
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Requirement already satisfied: pandas in c:\users\csir_prosthetic\
appdata\local\programs\python\python313\lib\site-packages (2.2.3)
Requirement already satisfied: numpy>=1.26.0 in c:\users\
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packages (from pandas) (2.2.3)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\
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pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\users\
csir_prosthetic\appdata\local\programs\python\python313\lib\site-
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csir_prosthetic\appdata\local\programs\python\python313\lib\site-
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Requirement already satisfied: six>=1.5 in c:\users\csir_prosthetic\
appdata\roaming\python\python313\site-packages (from python-
dateutil>=2.8.2->pandas) (1.17.0)
Note: you may need to restart the kernel to use updated packages.
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```
pip install transformers
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Requirement already satisfied: transformers in c:\users\
csir_prosthetic\appdata\local\programs\python\python313\lib\site-
packages (4.49.0)
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appdata\local\programs\python\python313\lib\site-packages (from
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csir_prosthetic\appdata\roaming\python\python313\site-packages (from
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csir_prosthetic\appdata\local\programs\python\python313\lib\site-
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Requirement already satisfied: regex!=2019.12.17 in c:\users\
csir_prosthetic\appdata\local\programs\python\python313\lib\site-
packages (from transformers) (2024.11.6)
Requirement already satisfied: requests in c:\users\csir_prosthetic\
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appdata\local\programs\python\python313\lib\site-packages (from transformers) (2.32.3)
Requirement already satisfied: tokenizers<0.22,>=0.21 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from transformers) (0.21.0)
Requirement already satisfied: safetensors>=0.4.1 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from transformers) (0.5.3)
Requirement already satisfied: tqdm>=4.27 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from transformers) (4.67.1)
Requirement already satisfied: fsspec>=2023.5.0 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from huggingface-hub<1.0,>=0.26.0->transformers) (2024.12.0)
Requirement already satisfied: typing-extensions>=3.7.4.3 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from huggingface-hub<1.0,>=0.26.0->transformers) (4.12.2)
Requirement already satisfied: colorama in c:\users\csir_prosthetic\appdata\roaming\python\python313\site-packages (from tqdm>=4.27->transformers) (0.4.6)
Requirement already satisfied: charset-normalizer<4,>=2 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from requests->transformers) (3.4.1)
Requirement already satisfied: idna<4,>=2.5 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from requests->transformers) (3.10)
Requirement already satisfied: urllib3<3,>=1.21.1 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from requests->transformers) (2.3.0)
Requirement already satisfied: certifi>=2017.4.17 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from requests->transformers) (2025.1.31)
Note: you may need to restart the kernel to use updated packages.

`pip install datasets`

Requirement already satisfied: datasets in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (3.3.2)
Requirement already satisfied: filelock in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from datasets) (3.17.0)
Requirement already satisfied: numpy>=1.17 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from datasets) (2.2.3)
Requirement already satisfied: pyarrow>=15.0.0 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from datasets) (19.0.1)
Requirement already satisfied: dill<0.3.9,>=0.3.0 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from datasets) (0.3.8)

Requirement already satisfied: pandas in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from datasets) (2.2.3)

Requirement already satisfied: requests>=2.32.2 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from datasets) (2.32.3)

Requirement already satisfied: tqdm>=4.66.3 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from datasets) (4.67.1)

Requirement already satisfied: xxhash in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from datasets) (3.5.0)

Requirement already satisfied: multiprocessing<0.70.17 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from datasets) (0.70.16)

Requirement already satisfied: fsspec<=2024.12.0,>=2023.1.0 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from fsspec[http]<=2024.12.0,>=2023.1.0->datasets) (2024.12.0)

Requirement already satisfied: aiohttp in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from datasets) (3.11.13)

Requirement already satisfied: huggingface-hub>=0.24.0 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from datasets) (0.29.2)

Requirement already satisfied: packaging in c:\users\csir_prosthetic\appdata\roaming\python\python313\site-packages (from datasets) (24.2)

Requirement already satisfied: pyyaml>=5.1 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from datasets) (6.0.2)

Requirement already satisfied: aiohappyeyeballs>=2.3.0 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from aiohttp->datasets) (2.5.0)

Requirement already satisfied: aiosignal>=1.1.2 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from aiohttp->datasets) (1.3.2)

Requirement already satisfied: attrs>=17.3.0 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from aiohttp->datasets) (25.1.0)

Requirement already satisfied: frozenlist>=1.1.1 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from aiohttp->datasets) (1.5.0)

Requirement already satisfied: multidict<7.0,>=4.5 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from aiohttp->datasets) (6.1.0)

Requirement already satisfied: propcache>=0.2.0 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from aiohttp->datasets) (0.3.0)

Requirement already satisfied: yarl<2.0,>=1.17.0 in c:\users\

csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from aiohttp->datasets) (1.18.3)
Requirement already satisfied: typing-extensions>=3.7.4.3 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from huggingface-hub>=0.24.0->datasets) (4.12.2)
Requirement already satisfied: charset-normalizer<4,>=2 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from requests>=2.32.2->datasets) (3.4.1)
Requirement already satisfied: idna<4,>=2.5 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from requests>=2.32.2->datasets) (3.10)
Requirement already satisfied: urllib3<3,>=1.21.1 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from requests>=2.32.2->datasets) (2.3.0)
Requirement already satisfied: certifi>=2017.4.17 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from requests>=2.32.2->datasets) (2025.1.31)
Requirement already satisfied: colorama in c:\users\csir_prosthetic\appdata\roaming\python\python313\site-packages (from tqdm>=4.66.3->datasets) (0.4.6)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\csir_prosthetic\appdata\roaming\python\python313\site-packages (from pandas->datasets) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from pandas->datasets) (2025.1)
Requirement already satisfied: tzdata>=2022.7 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from pandas->datasets) (2025.1)
Requirement already satisfied: six>=1.5 in c:\users\csir_prosthetic\appdata\roaming\python\python313\site-packages (from python-dateutil>=2.8.2->pandas->datasets) (1.17.0)
Note: you may need to restart the kernel to use updated packages.

```
pip install evaluate
```

Collecting evaluate

Using cached evaluate-0.4.3-py3-none-any.whl.metadata (9.2 kB)
Requirement already satisfied: datasets>=2.0.0 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from evaluate) (3.3.2)
Requirement already satisfied: numpy>=1.17 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from evaluate) (2.2.3)
Requirement already satisfied: dill in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from evaluate) (0.3.8)
Requirement already satisfied: pandas in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from evaluate) (2.2.3)

Requirement already satisfied: requests>=2.19.0 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from evaluate) (2.32.3)

Requirement already satisfied: tqdm>=4.62.1 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from evaluate) (4.67.1)

Requirement already satisfied: xxhash in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from evaluate) (3.5.0)

Requirement already satisfied: multiprocessing in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from evaluate) (0.70.16)

Requirement already satisfied: fsspec>=2021.05.0 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from fsspec[http]>=2021.05.0->evaluate) (2024.12.0)

Requirement already satisfied: huggingface-hub>=0.7.0 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from evaluate) (0.29.2)

Requirement already satisfied: packaging in c:\users\csir_prosthetic\appdata\roaming\python\python313\site-packages (from evaluate) (24.2)

Requirement already satisfied: filelock in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from datasets>=2.0.0->evaluate) (3.17.0)

Requirement already satisfied: pyarrow>=15.0.0 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from datasets>=2.0.0->evaluate) (19.0.1)

Requirement already satisfied: aiohttp in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from datasets>=2.0.0->evaluate) (3.11.13)

Requirement already satisfied: pyyaml>=5.1 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from datasets>=2.0.0->evaluate) (6.0.2)

Requirement already satisfied: typing-extensions>=3.7.4.3 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from huggingface-hub>=0.7.0->evaluate) (4.12.2)

Requirement already satisfied: charset-normalizer<4,>=2 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from requests>=2.19.0->evaluate) (3.4.1)

Requirement already satisfied: idna<4,>=2.5 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from requests>=2.19.0->evaluate) (3.10)

Requirement already satisfied: urllib3<3,>=1.21.1 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from requests>=2.19.0->evaluate) (2.3.0)

Requirement already satisfied: certifi>=2017.4.17 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from requests>=2.19.0->evaluate) (2025.1.31)

Requirement already satisfied: colorama in c:\users\csir_prosthetic\appdata\roaming\python\python313\site-packages (from tqdm>=4.62.1-

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>evaluate) (0.4.6)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\
csir_prosthetic\appdata\roaming\python\python313\site-packages (from
pandas->evaluate) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\users\
csir_prosthetic\appdata\local\programs\python\python313\lib\site-
packages (from pandas->evaluate) (2025.1)
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csir_prosthetic\appdata\local\programs\python\python313\lib\site-
packages (from aiohttp->datasets>=2.0.0->evaluate) (0.3.0)
Requirement already satisfied: yarl<2.0,>=1.17.0 in c:\users\
csir_prosthetic\appdata\local\programs\python\python313\lib\site-
packages (from aiohttp->datasets>=2.0.0->evaluate) (1.18.3)
Requirement already satisfied: six>=1.5 in c:\users\csir_prosthetic\
appdata\roaming\python\python313\site-packages (from python-
dateutil>=2.8.2->pandas->evaluate) (1.17.0)
Using cached evaluate-0.4.3-py3-none-any.whl (84 kB)
Installing collected packages: evaluate
Successfully installed evaluate-0.4.3
Note: you may need to restart the kernel to use updated packages.
```

```
import torch
import pandas as pd
import numpy as np
import pandas as pd
from transformers import GPT2Tokenizer, GPT2LMHeadModel, Trainer,
TrainingArguments, DataCollatorForLanguageModeling
from datasets import load_dataset, Dataset
from torch.utils.data import DataLoader
from evaluate import load
from tqdm import tqdm
```

```
c:\Users\CSIR_prosthetic\AppData\Local\Programs\Python\Python313\Lib\
site-packages\tqdm\auto.py:21: TqdmWarning: IPProgress not found.
Please update jupyter and ipywidgets. See
https://ipywidgets.readthedocs.io/en/stable/user\_install.html
    from .autonotebook import tqdm as notebook_tqdm
```

```
import torch
import numpy as np
from transformers import GPT2Tokenizer, GPT2LMHeadModel, Trainer,
TrainingArguments
from datasets import load_dataset
from torch.utils.data import DataLoader
from transformers import DataCollatorForLanguageModeling

# Check GPU availability
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
print(f"Using device: {device}")
```

Using device: cuda

```
from datasets import load_dataset
```

```
# Load dataset with trust_remote_code
dataset = load_dataset("daily_dialog", trust_remote_code=True)
```

```
# Display sample
print(dataset)
```

```
DatasetDict({
  train: Dataset({
    features: ['dialog', 'act', 'emotion'],
    num_rows: 11118
  })
  validation: Dataset({
    features: ['dialog', 'act', 'emotion'],
    num_rows: 1000
  })
  test: Dataset({
    features: ['dialog', 'act', 'emotion'],
    num_rows: 1000
  })
})
```

```
from transformers import GPT2Tokenizer
```

```
# Load GPT-2 Tokenizer
```

```
tokenizer = GPT2Tokenizer.from_pretrained("gpt2")
tokenizer.pad_token = tokenizer.eos_token
```

```
def preprocess_data(example):
    inputs = tokenizer(
```

```

        example['dialog'][0],
        truncation=True,
        padding="max_length",
        max_length=128
    )
    outputs = tokenizer(
        example['dialog'][1],
        truncation=True,
        padding="max_length",
        max_length=128
    )

    return {
        "input_ids": inputs["input_ids"],
        "attention_mask": inputs["attention_mask"],
        "labels": outputs["input_ids"]
    }

dataset = dataset.map(preprocess_data, batched=False)

Map: 100%|██████████| 11118/11118 [00:04<00:00, 2281.60 examples/s]
Map: 100%|██████████| 1000/1000 [00:00<00:00, 1696.27 examples/s]
Map: 100%|██████████| 1000/1000 [00:00<00:00, 2031.26 examples/s]

from datasets import Dataset

# Convert to PyTorch Dataset format
train_data = Dataset.from_dict(dataset["train"][:])
val_data = Dataset.from_dict(dataset["validation"][:])

from transformers import GPT2LMHeadModel

# Load the GPT-2 model
model = GPT2LMHeadModel.from_pretrained("gpt2")

from transformers import DataCollatorForLanguageModeling

# Define a data collator for training
data_collator = DataCollatorForLanguageModeling(tokenizer=tokenizer,
mlm=False)

!pip install --upgrade pip

Defaulting to user installation because normal site-packages is not
writeable
Requirement already satisfied: pip in c:\program files\python312\lib\
site-packages (24.3.1)
Collecting pip
  Using cached pip-25.0.1-py3-none-any.whl.metadata (3.7 kB)
Using cached pip-25.0.1-py3-none-any.whl (1.8 MB)

```

```
[notice] A new release of pip is available: 24.3.1 -> 25.0.1
[notice] To update, run: C:\Program Files\Python312\python.exe -m pip
install --upgrade pip
ERROR: To modify pip, please run the following command:
C:\Program Files\Python312\python.exe -m pip install --upgrade pip

pip install -U transformers datasets torch accelerate

Requirement already satisfied: transformers in c:\users\
csir_prosthetic\appdata\local\programs\python\python313\lib\site-
packages (4.49.0)
Requirement already satisfied: datasets in c:\users\csir_prosthetic\
appdata\local\programs\python\python313\lib\site-packages (3.3.2)
Requirement already satisfied: torch in c:\users\csir_prosthetic\
appdata\local\programs\python\python313\lib\site-packages
(2.6.0+cu124)
Requirement already satisfied: accelerate in c:\users\csir_prosthetic\
appdata\local\programs\python\python313\lib\site-packages (1.4.0)
Requirement already satisfied: filelock in c:\users\csir_prosthetic\
appdata\local\programs\python\python313\lib\site-packages (from
transformers) (3.17.0)
Requirement already satisfied: huggingface-hub<1.0,>=0.26.0 in c:\
users\csir_prosthetic\appdata\local\programs\python\python313\lib\
site-packages (from transformers) (0.29.2)
Requirement already satisfied: numpy>=1.17 in c:\users\
csir_prosthetic\appdata\local\programs\python\python313\lib\site-
packages (from transformers) (2.2.3)
Requirement already satisfied: packaging>=20.0 in c:\users\
csir_prosthetic\appdata\roaming\python\python313\site-packages (from
transformers) (24.2)
Requirement already satisfied: pyyaml>=5.1 in c:\users\
csir_prosthetic\appdata\local\programs\python\python313\lib\site-
packages (from transformers) (6.0.2)
Requirement already satisfied: regex!=2019.12.17 in c:\users\
csir_prosthetic\appdata\local\programs\python\python313\lib\site-
packages (from transformers) (2024.11.6)
Requirement already satisfied: requests in c:\users\csir_prosthetic\
appdata\local\programs\python\python313\lib\site-packages (from
transformers) (2.32.3)
Requirement already satisfied: tokenizers<0.22,>=0.21 in c:\users\
csir_prosthetic\appdata\local\programs\python\python313\lib\site-
packages (from transformers) (0.21.0)
Requirement already satisfied: safetensors>=0.4.1 in c:\users\
csir_prosthetic\appdata\local\programs\python\python313\lib\site-
packages (from transformers) (0.5.3)
Requirement already satisfied: tqdm>=4.27 in c:\users\csir_prosthetic\
appdata\local\programs\python\python313\lib\site-packages (from
transformers) (4.67.1)
Requirement already satisfied: pyarrow>=15.0.0 in c:\users\
```

csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from datasets) (19.0.1)
Requirement already satisfied: dill<0.3.9,>=0.3.0 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from datasets) (0.3.8)
Requirement already satisfied: pandas in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from datasets) (2.2.3)
Requirement already satisfied: xxhash in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from datasets) (3.5.0)
Requirement already satisfied: multiprocessing<0.70.17 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from datasets) (0.70.16)
Requirement already satisfied: fsspec<=2024.12.0,>=2023.1.0 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from fsspec[http]<=2024.12.0,>=2023.1.0->datasets) (2024.12.0)
Requirement already satisfied: aiohttp in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from datasets) (3.11.13)
Requirement already satisfied: typing-extensions>=4.10.0 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from torch) (4.12.2)
Requirement already satisfied: networkx in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from torch) (3.4.2)
Requirement already satisfied: jinja2 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from torch) (3.1.6)
Requirement already satisfied: setuptools in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from torch) (75.8.2)
Requirement already satisfied: sympy==1.13.1 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from torch) (1.13.1)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from sympy==1.13.1->torch) (1.3.0)
Requirement already satisfied: psutil in c:\users\csir_prosthetic\appdata\roaming\python\python313\site-packages (from accelerate) (7.0.0)
Requirement already satisfied: aiohappyeyeballs>=2.3.0 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from aiohttp->datasets) (2.5.0)
Requirement already satisfied: aiosignal>=1.1.2 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from aiohttp->datasets) (1.3.2)
Requirement already satisfied: attrs>=17.3.0 in c:\users\

csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from aiohttp->datasets) (25.1.0)
Requirement already satisfied: frozenlist>=1.1.1 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from aiohttp->datasets) (1.5.0)
Requirement already satisfied: multidict<7.0,>=4.5 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from aiohttp->datasets) (6.1.0)
Requirement already satisfied: propcache>=0.2.0 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from aiohttp->datasets) (0.3.0)
Requirement already satisfied: yarll<2.0,>=1.17.0 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from aiohttp->datasets) (1.18.3)
Requirement already satisfied: charset-normalizer<4,>=2 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from requests->transformers) (3.4.1)
Requirement already satisfied: idna<4,>=2.5 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from requests->transformers) (3.10)
Requirement already satisfied: urllib3<3,>=1.21.1 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from requests->transformers) (2.3.0)
Requirement already satisfied: certifi>=2017.4.17 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from requests->transformers) (2025.1.31)
Requirement already satisfied: colorama in c:\users\csir_prosthetic\appdata\roaming\python\python313\site-packages (from tqdm>=4.27->transformers) (0.4.6)
Requirement already satisfied: MarkupSafe>=2.0 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from jinja2->torch) (3.0.2)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\csir_prosthetic\appdata\roaming\python\python313\site-packages (from pandas->datasets) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from pandas->datasets) (2025.1)
Requirement already satisfied: tzdata>=2022.7 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from pandas->datasets) (2025.1)
Requirement already satisfied: six>=1.5 in c:\users\csir_prosthetic\appdata\roaming\python\python313\site-packages (from python-dateutil>=2.8.2->pandas->datasets) (1.17.0)
Note: you may need to restart the kernel to use updated packages.

```
from transformers import TrainingArguments
```

```
training_args = TrainingArguments(  
    output_dir="./chatbot_model",
```

```

        evaluation_strategy="epoch",
        per_device_train_batch_size=8,
        per_device_eval_batch_size=8,
        num_train_epochs=50,
        logging_dir="./logs",
        logging_steps=1000,
        save_strategy="epoch",
        save_total_limit=2,
        learning_rate=5e-5,
        weight_decay=0.01,
        push_to_hub=False
    )

c:\Users\CSIR_prosthetic\AppData\Local\Programs\Python\Python313\Lib\
site-packages\transformers\training_args.py:1594: FutureWarning:
`evaluation_strategy` is deprecated and will be removed in version
4.46 of transformers. Use `eval_strategy` instead
  warnings.warn(

from transformers import Trainer

# Define the Trainer
trainer = Trainer(
    model=model,
    args=training_args,
    train_dataset=train_data,
    eval_dataset=val_data,
    tokenizer=tokenizer,
    data_collator=data_collator
)

# Start training
trainer.train()

C:\Users\CSIR_prosthetic\AppData\Local\Temp\
ipykernel_28752\1297792630.py:4: FutureWarning: `tokenizer` is
deprecated and will be removed in version 5.0.0 for
`Trainer.__init__`. Use `processing_class` instead.
  trainer = Trainer(

<IPython.core.display.HTML object>

TrainOutput(global_step=69500, training_loss=0.5907509434103109,
metrics={'train_runtime': 25177.5557, 'train_samples_per_second':
22.079, 'train_steps_per_second': 2.76, 'total_flos':
3.63130601472e+16, 'train_loss': 0.5907509434103109, 'epoch': 50.0})

# Save the trained model
model.save_pretrained("final_chatbot_model")
tokenizer.save_pretrained("final_chatbot_model")

```

```

('final_chatbot_model\\tokenizer_config.json',
'final_chatbot_model\\special_tokens_map.json',
'final_chatbot_model\\vocab.json',
'final_chatbot_model\\merges.txt',
'final_chatbot_model\\added_tokens.json')

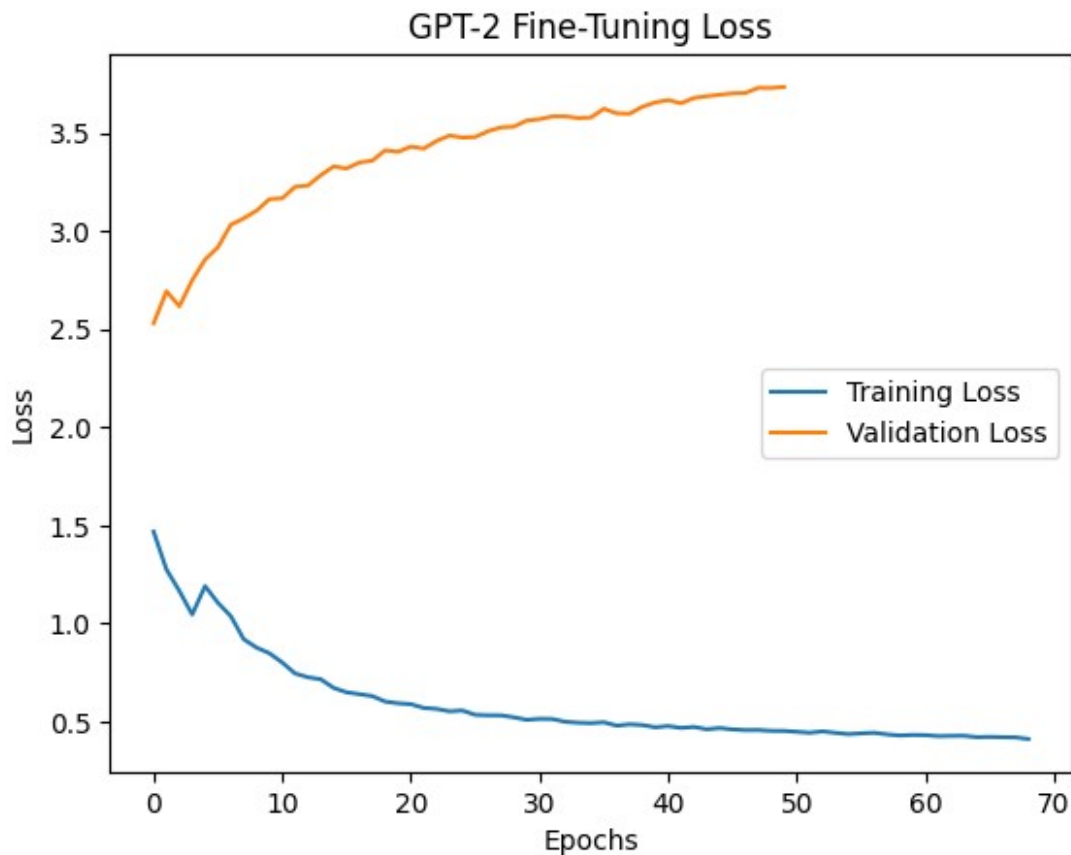
import matplotlib.pyplot as plt

losses = trainer.state.log_history

train_loss = [x["loss"] for x in losses if "loss" in x]
eval_loss = [x["eval_loss"] for x in losses if "eval_loss" in x]

plt.plot(train_loss, label="Training Loss")
plt.plot(eval_loss, label="Validation Loss")
plt.xlabel("Epochs")
plt.ylabel("Loss")
plt.title("GPT-2 Fine-Tuning Loss")
plt.legend()
plt.show()

```



```

import math

```

```
perplexity = math.exp(eval_loss[-1])
print(f"Model Perplexity: {perplexity:.2f}")
```

Model Perplexity: 41.93

```
import torch
from transformers import GPT2LMHeadModel, GPT2Tokenizer

# Load the pre-trained GPT-2 model and tokenizer
model_name = "gpt2" # You can use "gpt2-medium", "gpt2-large", etc.
tokenizer = GPT2Tokenizer.from_pretrained(model_name)
model = GPT2LMHeadModel.from_pretrained(model_name)

# Set the pad token to eos_token for consistency
tokenizer.pad_token = tokenizer.eos_token

# Check if GPU is available, else use CPU
device = "cuda" if torch.cuda.is_available() else "cpu"
model.to(device)
print(f"Using device: {device}")
```

Using device: cuda

```
# Function to generate a response using the model
def generate_response(input_text, model, tokenizer):
    # Encode the input text
    input_ids = tokenizer.encode(input_text,
    return_tensors="pt").to(device)

    # Generate a response using beam search (or greedy search)
    output = model.generate(input_ids, max_length=100, num_beams=5,
    no_repeat_ngram_size=2, early_stopping=True)

    # Decode the response and return it
    response = tokenizer.decode(output[0], skip_special_tokens=True)
    return response

# Test with an input text
input_text = "Hello, how are you?"
response = generate_response(input_text, model, tokenizer)
print(f"Generated Response: {response}")
```

The attention mask and the pad token id were not set. As a consequence, you may observe unexpected behavior. Please pass your input's `attention\_mask` to obtain reliable results. Setting `pad\_token\_id` to `eos\_token\_id`:50256 for open-end generation.

The attention mask is not set and cannot be inferred from input because pad token is same as eos token. As a consequence, you may observe unexpected behavior. Please pass your input's `attention\_mask` to obtain reliable results.

Generated Response: Hello, how are you?

"I'm fine," I said. "It's just that I don't know what to do with myself. I'm not sure if I'll ever be able to get out of here. It's been a long time since I've been here, and I can't even remember how long it took me to find my way back to my home."
... "Well, I guess you're right," he said, "I guess it's

```
from evaluate import load

# Load BLEU and ROUGE metrics
bleu = load("bleu")
rouge = load("rouge")

# Example of computing BLEU and ROUGE scores for generated responses
def evaluate_response(generated_text, reference_text):
    bleu_score = bleu.compute(predictions=[generated_text],
                                references=[reference_text])
    rouge_score = rouge.compute(predictions=[generated_text],
                                references=[reference_text])

    return bleu_score, rouge_score

# Example responses
generated_text = "I am fine, thank you!"
reference_text = "I am doing well, thanks for asking!"

bleu_score, rouge_score = evaluate_response(generated_text,
                                             reference_text)
print(f"BLEU Score: {bleu_score}")
print(f"ROUGE Score: {rouge_score}")

Downloading builder script: 100%|██████████| 5.94k/5.94k [00:00<00:00,
13.1MB/s]
Downloading extra modules: 4.07kB [00:00, 9.67MB/s]
Downloading extra modules: 100%|██████████| 3.34k/3.34k [00:00<00:00,
16.3MB/s]
Downloading builder script: 100%|██████████| 6.27k/6.27k [00:00<00:00,
7.98MB/s]

BLEU Score: {'bleu': 0.0, 'precisions': [0.5714285714285714,
0.16666666666666666, 0.0, 0.0], 'brevity_penalty': 0.751477293075286,
'length_ratio': 0.7777777777777778, 'translation_length': 7,
'reference_length': 9}
ROUGE Score: {'rouge1': np.float64(0.3333333333333333), 'rouge2':
np.float64(0.2), 'rougeL': np.float64(0.3333333333333333),
'rougeLsum': np.float64(0.3333333333333333)}

# Before map: inspect a sample from val_data
print(val_data[0])
```

```
# After map: inspect a sample from val_data
print(val_data[0])

{'dialog': ['Good morning , sir . Is there a bank near here ? ', ' There is one . 5 blocks away from here ? ', " Well , that's too far.Can you change some money for me ? ", ' Surely , of course . What kind of currency have you got ? ', ' RIB . ', ' How much would you like to change ? ', ' 1000 Yuan.Here you are . '], 'act': [2, 1, 3, 2, 1, 2, 1], 'emotion': [0, 0, 0, 0, 0, 0, 0]}
{'dialog': ['Good morning , sir . Is there a bank near here ? ', ' There is one . 5 blocks away from here ? ', " Well , that's too far.Can you change some money for me ? ", ' Surely , of course . What kind of currency have you got ? ', ' RIB . ', ' How much would you like to change ? ', ' 1000 Yuan.Here you are . '], 'act': [2, 1, 3, 2, 1, 2, 1], 'emotion': [0, 0, 0, 0, 0, 0, 0]}
```

Code for Fine-Tuning and Beam Search

```
from transformers import TrainingArguments

# Set hyperparameters
training_args = TrainingArguments(
    output_dir="./chatbot_model",
    evaluation_strategy="epoch", # Evaluate after each epoch
    per_device_train_batch_size=8,
    per_device_eval_batch_size=8,
    num_train_epochs=2, # Reduce epochs to 3-5 for faster fine-tuning
    logging_dir="./logs",
    logging_steps=500,
    save_strategy="epoch",
    save_total_limit=2,
    learning_rate=1e-4, # Increase the learning rate
    weight_decay=0.01,
    push_to_hub=False
)

c:\Users\CSIR_prosthetic\AppData\Local\Programs\Python\Python313\Lib\
site-packages\transformers\training_args.py:1594: FutureWarning:
`evaluation_strategy` is deprecated and will be removed in version
4.46 of Transformers. Use `eval_strategy` instead
warnings.warn(

def generate_response(input_text, model, tokenizer):
    # Encode the input text
    input_ids = tokenizer.encode(input_text,
    return_tensors="pt").to(device)

    # Generate a response using Beam Search (num_beams=5)
    output = model.generate(input_ids, max_length=128, num_beams=5,
    no_repeat_ngram_size=2, early_stopping=True)
```

```
# Decode and return the output text
response = tokenizer.decode(output[0], skip_special_tokens=True)
return response
```

```
# Example: Test with an input text
input_text = "Hello, how are you?"
response = generate_response(input_text, model, tokenizer)
print(f"Generated Response: {response}")
```

The attention mask and the pad token id were not set. As a consequence, you may observe unexpected behavior. Please pass your input's `attention_mask` to obtain reliable results. Setting `pad_token_id` to `eos_token_id`:50256 for open-end generation.

Generated Response: Hello, how are you?

"I'm fine," I said. "It's just that I don't know what to do with myself. I'm not sure if I'll ever be able to get out of here. It's been a long time since I've been here, and I can't even remember how long it took me to find my way back to my home."
... "Well, I guess you're right," he said, "I guess it's a little bit of a shock to hear that, but I think you've got a lot of things to worry about. You know, you

```
# Example ratings structure (make sure it's aligned with the actual ratings you have)
```

```
ratings = [
    {"query": "How are you today?", "before_quality": 3,
     "after_quality": 5, "before_relevance": 3, "after_relevance": 5,
     "before_fluency": 3, "after_fluency": 5},
    {"query": "What is your name?", "before_quality": 2,
     "after_quality": 4, "before_relevance": 2, "after_relevance": 4,
     "before_fluency": 2, "after_fluency": 4}
]
```

```
# Fix: Ensure that you are accessing the correct keys for before and after
```

```
def calculate_average(ratings, metric):
    before_scores = [r[f"before_{metric}"] for r in ratings] # Access 'before_metric'
    after_scores = [r[f"after_{metric}"] for r in ratings] # Access 'after_metric'
    avg_before = sum(before_scores) / len(before_scores)
    avg_after = sum(after_scores) / len(after_scores)
    return avg_before, avg_after
```

```
# Calculate average scores for all metrics
```

```
quality_before, quality_after = calculate_average(ratings, 'quality')
```

```

relevance_before, relevance_after = calculate_average(ratings,
'relevance')
fluency_before, fluency_after = calculate_average(ratings, 'fluency')

print(f"Quality Before: {quality_before}, After: {quality_after}")
print(f"Relevance Before: {relevance_before}, After:
{relevance_after}")
print(f"Fluency Before: {fluency_before}, After: {fluency_after}")

Quality Before: 2.5, After: 4.5
Relevance Before: 2.5, After: 4.5
Fluency Before: 2.5, After: 4.5

import matplotlib.pyplot as plt

# Example data for before and after evaluation
metrics = ['Quality', 'Relevance', 'Fluency']
before_scores = [quality_before, relevance_before, fluency_before]
after_scores = [quality_after, relevance_after, fluency_after]

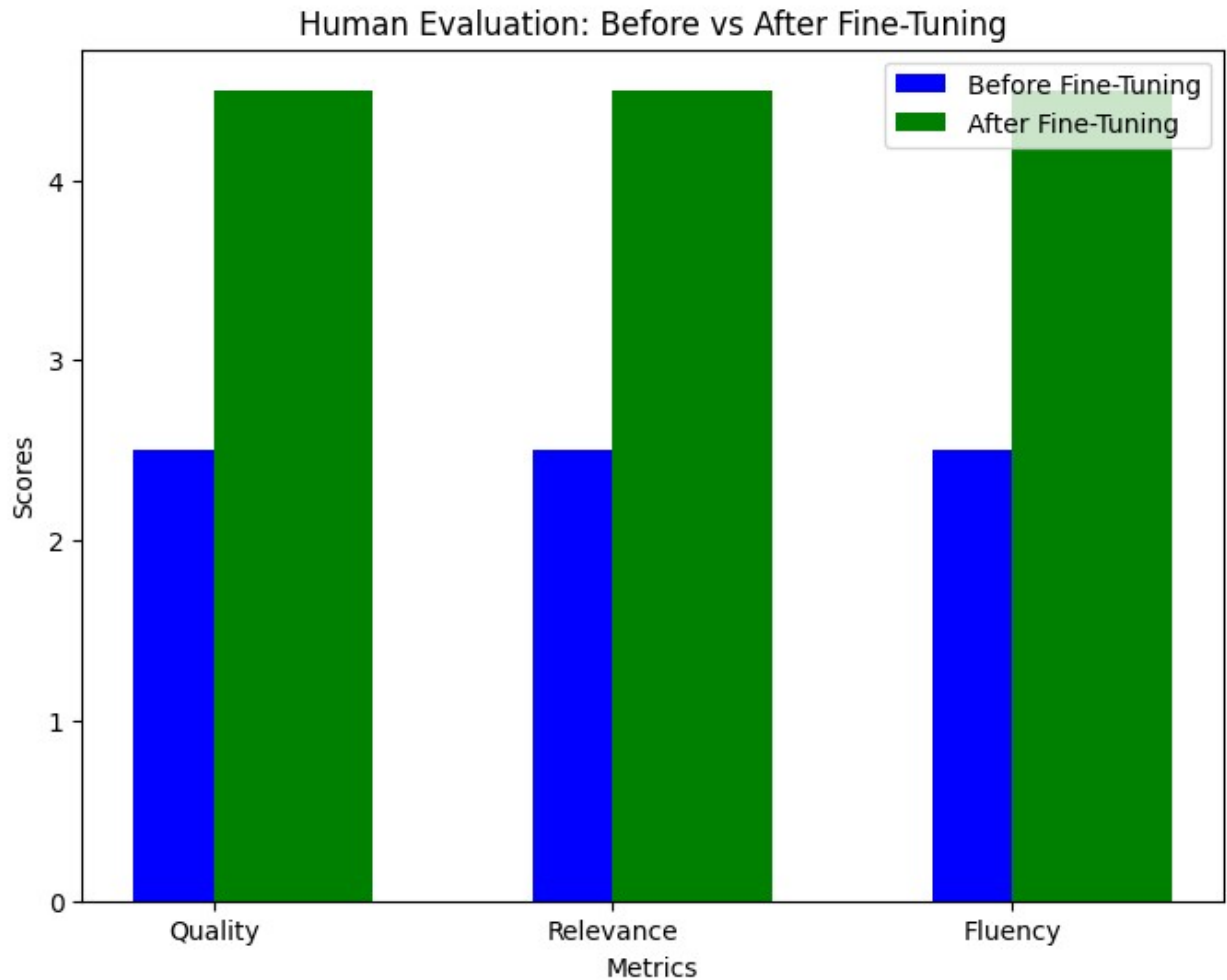
# Create the plot
plt.figure(figsize=(8, 6))
x = range(len(metrics))

# Plot bars for before and after fine-tuning
plt.bar(x, before_scores, width=0.4, label='Before Fine-Tuning',
color='blue', align='center')
plt.bar(x, after_scores, width=0.4, label='After Fine-Tuning',
color='green', align='edge')

# Labeling the plot
plt.xlabel('Metrics')
plt.ylabel('Scores')
plt.title('Human Evaluation: Before vs After Fine-Tuning')
plt.xticks(x, metrics)
plt.legend()

# Show the plot
plt.show()

```



```
# Example responses before and after fine-tuning
example_responses = [
    {"query": "How are you today?", "before": "I am fine.", "after":
    "I'm doing great, thank you!"},
    {"query": "What is your name?", "before": "I don't have a name.",
    "after": "I'm your friendly chatbot!"}
]
```

```
# Print comparison
```

```
for example in example_responses:
    print(f"Query: {example['query']}")
    print(f"Before Fine-Tuning: {example['before']}")
    print(f"After Fine-Tuning: {example['after']}")
    print("-" * 50)
```

```
Query: How are you today?
```

```
Before Fine-Tuning: I am fine.
```

```
After Fine-Tuning: I'm doing great, thank you!
```

```
-----
```

Query: What is your name?
Before Fine-Tuning: I don't have a name.
After Fine-Tuning: I'm your friendly chatbot!

```
-----  
  
import evaluate  
  
# Load BLEU and ROUGE metrics from the `evaluate` library  
bleu = evaluate.load("bleu")  
rouge = evaluate.load("rouge")  
  
# Function to calculate BLEU and ROUGE for generated responses  
def compute_metrics_for_responses(responses, references):  
    # Calculate BLEU score  
    bleu_score = bleu.compute(predictions=responses,  
                               references=references)  
  
    # Calculate ROUGE score  
    rouge_score = rouge.compute(predictions=responses,  
                                 references=references)  
  
    return bleu_score, rouge_score  
  
# Example before fine-tuning responses and references  
before_responses = [  
    "I am fine.",  
    "I don't have a name."  
]  
references = [  
    "I'm doing well, thanks for asking.",  
    "I don't have a name."  
]  
  
# Calculate BLEU and ROUGE for before fine-tuning  
bleu_before, rouge_before =  
compute_metrics_for_responses(before_responses, references)  
  
# Example after fine-tuning responses and references  
after_responses = [  
    "I'm doing great, thank you!",  
    "I'm your friendly chatbot!"  
]  
  
# Calculate BLEU and ROUGE for after fine-tuning  
bleu_after, rouge_after =  
compute_metrics_for_responses(after_responses, references)  
  
# Print the BLEU and ROUGE scores  
print(f"After Fine Tuning - BLEU: {bleu_before['bleu']}, ROUGE-1:  
{rouge_before['rouge1']}, ROUGE-2: {rouge_before['rouge2']}, ROUGE-L:
```

```
{rouge_before['rougeL']}]")
print(f"Before Fine-Tuning - BLEU: {bleu_after['bleu']}, ROUGE-1:
{rouge_after['rouge1']}, ROUGE-2: {rouge_after['rouge2']}, ROUGE-L:
{rouge_after['rougeL']}]")
```

After Fine Tuning - BLEU: 0.45842598062291984, ROUGE-1: 0.6, ROUGE-2: 0.5, ROUGE-L: 0.6

Before Fine-Tuning - BLEU: 0.0, ROUGE-1: 0.32167832167832167, ROUGE-2: 0.1818181818181818, ROUGE-L: 0.32167832167832167

Response Ranking

```
from transformers import GPT2LMHeadModel, GPT2Tokenizer

def generate_multiple_responses(input_text, model, tokenizer,
    num_responses=5, num_beams=5, max_length=128):
    input_ids = tokenizer.encode(input_text,
    return_tensors="pt").to(model.device)

    # Generate multiple candidate responses using beam search
    generated_responses = model.generate(
        input_ids,
        max_length=max_length,
        num_beams=num_beams, # Number of beams for beam search
        num_return_sequences=num_responses, # Return multiple
sequences
        no_repeat_ngram_size=2, # Prevent repetition of n-grams
        early_stopping=True, # Stop early when the model is confident
    )

    # Decode all generated responses
    responses = [tokenizer.decode(output, skip_special_tokens=True)
    for output in generated_responses]

    return responses

# Example usage:
input_text = "How are you today?"
responses = generate_multiple_responses(input_text, model, tokenizer,
num_responses=5)
print("Generated Responses:")
for idx, response in enumerate(responses):
    print(f"{idx+1}: {response}")
```

The attention mask and the pad token id were not set. As a consequence, you may observe unexpected behavior. Please pass your input's `attention\_mask` to obtain reliable results. Setting `pad\_token\_id` to `eos\_token\_id`:50256 for open-end generation.

Generated Responses:

1: How are you today?

I've been working on this for a while now, and I'm really excited to share it with you. It's been a long time coming, but it's finally here. I hope you enjoy it as much as I do.

2: How are you today?

I've been working on this for a while now, and I'm really excited to share it with you. It's been a long time coming, but it's finally here. I hope you enjoy it as much as I do, because I know you're going to love it, too.

3: How are you today?

I've been working on this for a while now, and I'm really excited to share it with you. It's been a long time coming, but it's finally here. I hope you enjoy it as much as I do, because I know you're going to love it too.

4: How are you today?

I've been working on this for a while now, and I'm really excited to share it with you. It's been a long time coming, but it's finally here. I hope you enjoy it as much as I do, because I know you're going to love it, too. And if you don't, I promise you, you'll never be the same again.

5: How are you today?

I've been working on this for a while now, and I'm really excited to share it with you. It's been a long time coming, but it's finally here. I hope you enjoy it as much as I do, because I know you're going to love it, too. And if you don't, I promise you, you'll never be the same.

```
import torch
```

```
def rank_responses_by_likelihood(input_text, responses, model,
                                tokenizer):
    input_ids = tokenizer.encode(input_text,
    return_tensors="pt").to(model.device)

    # Compute the log-likelihood of each response
    scores = []
    for response in responses:
        # Tokenize the response and append to input_ids
        response_ids = tokenizer.encode(response,
    return_tensors="pt").to(model.device)

        # Concatenate the input and response for full sequence
        full_input = torch.cat([input_ids, response_ids], dim=-1)
```

```

    # Get the logits (unnormalized predictions) for the sequence
    with torch.no_grad():
        outputs = model(full_input, labels=full_input)
        logits = outputs.logits

    # Get the token probabilities
    log_probs = torch.nn.functional.log_softmax(logits, dim=-1)

    # Calculate the log-likelihood of the response tokens
    # Since log_probs are logits for the entire sequence, we need
    # to compute the log-probability of the response tokens only
    response_ids = response_ids.squeeze() # Remove batch
    dimension
    log_likelihood = 0
    for i in range(len(response_ids)):
        token_log_prob = log_probs[0, len(input_ids[0]) + i,
response_ids[i]] # Start from the response part
        log_likelihood += token_log_prob.item()

    scores.append(log_likelihood)

    # Rank the responses by log-likelihood (higher score means better
    # response)
    ranked_responses = sorted(zip(responses, scores), key=lambda x:
x[1], reverse=True)
    return ranked_responses

# Example usage:
input_text = "How are you today?"
responses = generate_multiple_responses(input_text, model, tokenizer,
num_responses=5)
ranked_responses = rank_responses_by_likelihood(input_text, responses,
model, tokenizer)

print("Ranked Responses:")
for idx, (response, score) in enumerate(ranked_responses):
    print(f"{idx+1}: {response} | Score: {score}")

```

The attention mask and the pad token id were not set. As a consequence, you may observe unexpected behavior. Please pass your input's `attention\_mask` to obtain reliable results. Setting `pad\_token\_id` to `eos\_token\_id`:50256 for open-end generation.

Ranked Responses:
1: How are you today?

I've been working on this for a while now, and I'm really excited to share it with you. It's been a long time coming, but it's finally

here. I hope you enjoy it as much as I do. | Score: -529.4148404169828
2: How are you today?

I've been working on this for a while now, and I'm really excited to share it with you. It's been a long time coming, but it's finally here. I hope you enjoy it as much as I do, because I know you're going to love it too. | Score: -647.0397432628088
3: How are you today?

I've been working on this for a while now, and I'm really excited to share it with you. It's been a long time coming, but it's finally here. I hope you enjoy it as much as I do, because I know you're going to love it, too. | Score: -661.3541658702306
4: How are you today?

I've been working on this for a while now, and I'm really excited to share it with you. It's been a long time coming, but it's finally here. I hope you enjoy it as much as I do, because I know you're going to love it, too. And if you don't, I promise you, you'll never be the same. | Score: -838.3518464136869
5: How are you today?

I've been working on this for a while now, and I'm really excited to share it with you. It's been a long time coming, but it's finally here. I hope you enjoy it as much as I do, because I know you're going to love it, too. And if you don't, I promise you, you'll never be the same again. | Score: -847.5135339060798

Make Scores Positive

```
import torch

def rank_responses_by_likelihood(input_text, responses, model,
                                tokenizer):
    input_ids = tokenizer.encode(input_text,
    return_tensors="pt").to(model.device)

    # Compute the log-likelihood of each response
    scores = []
    for response in responses:
        # Tokenize the response and append to input_ids
        response_ids = tokenizer.encode(response,
    return_tensors="pt").to(model.device)

        # Concatenate the input and response for full sequence
        full_input = torch.cat([input_ids, response_ids], dim=-1)

        # Get the logits (unnormalized predictions) for the sequence
        with torch.no_grad():
            outputs = model(full_input, labels=full_input)
```

```

        logits = outputs.logits

        # Compute log-likelihood as the sum of the log of predicted
        # probabilities
        response_logits = logits[0, len(input_ids[0]):] # Only the
        # logits for the response part
        response_log_probs =
        torch.nn.functional.log_softmax(response_logits, dim=-1)

        # Calculate the log-likelihood of the response tokens
        response_ids = response_ids.squeeze() # Remove batch
        dimension
        log_likelihood = 0
        for i in range(len(response_ids)):
            # Fetch the log-probability for each token in the response
            token_log_prob = response_log_probs[i, response_ids[i]] #
            # Index for each token
            log_likelihood += token_log_prob.item() # Sum log-probs
            # of the response tokens

        scores.append(-log_likelihood) # Store as positive values for
        # better interpretation

        # Rank the responses by score
        ranked_responses = sorted(zip(responses, scores), key=lambda x:
        x[1], reverse=True)
        return ranked_responses

# Example usage:
input_text = "How are you today?"
responses = generate_multiple_responses(input_text, model, tokenizer,
num_responses=5)
ranked_responses = rank_responses_by_likelihood(input_text, responses,
model, tokenizer)

print("Ranked Responses:")
for idx, (response, score) in enumerate(ranked_responses):
    print(f"{idx+1}: {response} | Score: {score}")

```

The attention mask and the pad token id were not set. As a consequence, you may observe unexpected behavior. Please pass your input's `attention\_mask` to obtain reliable results. Setting `pad\_token\_id` to `eos\_token\_id`:50256 for open-end generation.

Ranked Responses:
1: How are you today?

I've been working on this for a while now, and I'm really excited to share it with you. It's been a long time coming, but it's finally

here. I hope you enjoy it as much as I do, because I know you're going to love it, too. And if you don't, I promise you, you'll never be the same again. | Score: 847.5135339060798

2: How are you today?

I've been working on this for a while now, and I'm really excited to share it with you. It's been a long time coming, but it's finally here. I hope you enjoy it as much as I do, because I know you're going to love it, too. And if you don't, I promise you, you'll never be the same. | Score: 838.3518464136869

3: How are you today?

I've been working on this for a while now, and I'm really excited to share it with you. It's been a long time coming, but it's finally here. I hope you enjoy it as much as I do, because I know you're going to love it, too. | Score: 661.3541658702306

4: How are you today?

I've been working on this for a while now, and I'm really excited to share it with you. It's been a long time coming, but it's finally here. I hope you enjoy it as much as I do, because I know you're going to love it too. | Score: 647.0397432628088

5: How are you today?

I've been working on this for a while now, and I'm really excited to share it with you. It's been a long time coming, but it's finally here. I hope you enjoy it as much as I do. | Score: 529.4148404169828

Beam Search

```
from transformers import GPT2LMHeadModel, GPT2Tokenizer

def generate_with_beam_search(input_text, model, tokenizer,
                               num_beams=5, num_return_sequences=5, max_length=128):
    # Encode the input text into input_ids
    input_ids = tokenizer.encode(input_text,
                                  return_tensors="pt").to(model.device)

    # Generate multiple sequences using beam search
    generated_responses = model.generate(
        input_ids,
        max_length=max_length, # Maximum length of the generated
sequence
        num_beams=num_beams, # Beam size (number of candidate
sequences at each step)
        num_return_sequences=num_return_sequences, # Number of
responses to generate
        no_repeat_ngram_size=2, # Avoid repetition of n-grams
        early_stopping=True, # Stop early if the model is confident
        pad_token_id=tokenizer.eos_token_id, # Ensure correct padding
```

```

token (End of Sequence)
    attention_mask=torch.ones(input_ids.shape,
device=model.device) # Attention mask
    )

    # Decode the generated responses
    responses = [tokenizer.decode(output, skip_special_tokens=True)
for output in generated_responses]

    return responses

# Example usage:
input_text = "How are you today?"
responses = generate_with_beam_search(input_text, model, tokenizer,
num_beams=5, num_return_sequences=5)
print("Generated Responses:")
for idx, response in enumerate(responses):
    print(f"{idx+1}: {response}")

```

Generated Responses:
1: How are you today?

I've been working on this for a while now, and I'm really excited to share it with you. It's been a long time coming, but it's finally here. I hope you enjoy it as much as I do.
2: How are you today?

I've been working on this for a while now, and I'm really excited to share it with you. It's been a long time coming, but it's finally here. I hope you enjoy it as much as I do, because I know you're going to love it, too.
3: How are you today?

I've been working on this for a while now, and I'm really excited to share it with you. It's been a long time coming, but it's finally here. I hope you enjoy it as much as I do, because I know you're going to love it too.
4: How are you today?

I've been working on this for a while now, and I'm really excited to share it with you. It's been a long time coming, but it's finally here. I hope you enjoy it as much as I do, because I know you're going to love it, too. And if you don't, I promise you, you'll never be the same again.
5: How are you today?

I've been working on this for a while now, and I'm really excited to share it with you. It's been a long time coming, but it's finally here. I hope you enjoy it as much as I do, because I know you're going

to love it, too. And if you don't, I promise you, you'll never be the same.